AdvJava

1)Which are the components of java programming language?

➔i)Variable

ii)Methods

iii)Constructors

iv)Blocks

v)Classes

vi)Interface

vii)Abstract Class

2)Which are the concept use in java programming language?

➔i)Class

ii)Object

iii)Abstraction

iv)Polymorphism

v)Encapsulation

vi)Inheritance

3)Define stand-alone application?

➔The application which is installed in one computer and perform action in the same computer , is Known as stand-alone application or non-server application.

4)Which are the technology provide adv-java to develop server-based application?

➔i)JDBC

ii)Servlet

iii)JSP

5)What is JDBC?

➔JDBC stands for “Java Database Connectivity” and which is used to interact with database Product.

6)What is Servlet?

➔Servlet means “Server Program” and which is used to accept the request from user through web Browser and provide the response.

7)What is JSP?

➔JSP stands for “Java Server Page” and which is response from server application(Web Application)

8)Define Storage?

➔The memory location where the data is available for accessing is known as Storage.

9)Which are the types of Storage?

➔**i)Field Storage**(The memory generated to hold a single-data values,is known as Field Storage)

ii)Object Storage(The memory generated to hold “group-members” is known as Object Storage)

iii)File Storage(The smallest permanent storage of computer system , which is “controlled and managed” by operating system , is known as File Storage)

iv)Database Storage(The largest permanent storage of computer system which is installed from externally is known as Database Storage)

10)Define Object Reference?

➔The address where the object is created is known as Object Reference.

11)Define Object Reference Variable?

➔The non-primitive data type variable , which is holding object reference is known as Object Reference Variable.

12)List of object generated from core java.

➔i)User defined class objects

ii)String objects

iii)Wrapper class objects

iv)Array objects

v)Collection<E> objects

vi)Map<K,V> objects

vii)Enum<E> objects

13)Define Stream?

➔The Continuous flow of data(0's and 1's) is known as Stream.

14)DisAdvantages of File Storage.

➔**i)Data redundancy**(Same information will be duplicated in different files)

ii)Data inconsistency(Data can be inconsistent due to data redundancy)

iii)Difficulty in accessing data(Because the data is available in scattered and no quering process)

iv)Limited data sharing(Because the data is scattered form)

v)File system corruption(File system can be corrupted due to the fragmentation or matadata)

vi)Security problem(File system will have security problems)

15)Define 'driver'?

➔The small software program which establishes connection between two end-point is known as 'driver'.

16)Define JDBC driver?

➔The driver which establishes connection between java program and database product is known as JDBC driver.

17)Types of JDBC driver.

➔i)JDBC-ODBC bridge driver

ii)Native API driver

iii)Native Protocol driver

iv)Thin driver

18)Which are methods of 'Connection' interface.

➔i)createStatement()

ii)prepareStatement()

iii)prepareCall()

iv)getAutoCommit()

v)setAutoCommit()

vi)setSavePoint()

vii)releaseSavePoint()

ix)commit()

x)close()

19)Define getRef()?

➔This getRef() method internally generate implementation object for interface Test , because getRef() method internally holding 'Anonymous Local InnerClass as implementation of class.

20)Define getConnection()?

➔getConnection() method internally generate implementation object for Connection Interface , Because getConnection() method internally holding 'Anonymous Local InnerClass' as implementation class of 'Connection'.

21)What is JDBC Statement?

➔-JDBC Statement will specify the the action to be performed on Database product.

-These JDBC Statement are categorized into three types.

i)Statement

ii)PreparedStatement

iii)CollableStatement

22)Define Statement?

➔-'Statement' is an interface from java.sql package and Which is used to execute normal queries (Create,Insert,Update,Select,Delete) without IN-Parameter.

-Syntax:

Statement stm = con.createStatement();

-The following are two important methods of 'Statement':

i)executeQuery() – Used to execute Select-queries.

ii)executeUpdate() – Used to execute NonSelect-queries.

23)Steps to establish Communication to Database product.

➔i)Loading driver

ii)Creating Connection to Database product

iii)Preparing JDBC Statement

iv)Executing query

v)Close the Connection from Database product.

24)Define PreparedStatement?

➔-'PreparedStatement' is an interface from java.sql package and which is used to execute normal queries with IN-Parameter.

-Syntax:

PreparedStatement ps = con.preparedStatement("query-structure");

-The following are two important methods of PreparedStatement interface.

i)executeQuery() – used for Select queries.

ii)executeUpdate() – used for Non-Select queries.

25)Define 'ResultSet' in JDBC?

➔'ResultSet' is an interface from java.sql package and which is instantiated to hold result Generated from Select-queries.

26)Types of ResultSet – Object.

➔i)NonScrollable ResultSet Object:-

-In NonScrollable ResultSet Object , the cursor can be moved only in one direction , which means top - of – table – data to bottom – of – table – data.

-Syntax:-

PreparedStatement ps = con.preparedStatement("Select – query – structure");

Result rs = ps.executeQuery();

ii)Scrollable ResultSet Object:-

-In Scrollable ResultSet Object , the cursor can be moved in both direction , which means Forward and backward.

-Syntax:-

PreparedStatement ps = con.preparedStatement("Select – query – structure",type,mode);

ResultSet rs = ps.executeQuery();

27)Define 'type'?

➔"type" specifies the direction of cursor on ResultSet Objects.

-The following field from "ResultSet' specify the "type":

public static final int TYPE_FORWORD_ONLY;

public static final int TYPE_SCROLL_INSENSITIVE;

public static final int TYPE_SCROLL_SENSITIVE;

28)Define 'mode'?

➔"mode" specifies the action to be performed on ResultSet Object.

-The following fields from 'ResultSet' specify the "mode":

public static final int CONCUR_READ_ONLY;

public static final int CONCUR_UPDATABLE;

29)Which are the important methods used to control the cursor on ResultSet Object?

➔i)afterLast()

ii)beforeFirst()

iii)previous()

iv)next()

v)first()

vi)last()

vii)absolute(int)

viii)relative(int)

30)Define Stream?

➔The continuous flow of data is known as Stream.

31)Types of Stream.

➔**-Java language supports two types of streams**

i)Byte Stream

-The continuous flow of data in the form of 8-bits is known as Byte Stream or Binary Stream.

-Through Byte stream we can send all types of multimedia data, which means we can send Text,

Audio, Video, Image and Animation.

ii)Character Stream

-The continuous flow of data in the form of 16-bits is known as Charactrer Stream or Text Stream.

-Character Stream is preferable for Text data, and which is not preferable for Audio, Video, Image and Animation data.

32)Which of the following SQL-Types to record stream data.

➔**a)BLOB:-**

BLOB stands for "Binary Large Object" and which is used to store Byte Stream data or Binary Stream data.

b)CLOB:-

CLOB stands for "Character Large Object" and which is used to store Character Stream data.

33)Define 'FileInputStream'?

➔'FileInputStream' class is from java.io package and which is instantiated to link(open) the file to Read Byte Stream data(binary stream data)

-Syntax:-

FileInputStream fis = new FileInputStream("fPath & fName");

34)Define setBinaryStream() method?

➔**-setBinaryStream()-** method is from 'PreparedStatement' and which specify the following.

i)Stream – source

ii)Stream – source linked to para-index

iii)Stream – length(file – size)

-Syntax:-

Ps.setBinaryStream(3 , fis , f.length());

35)Define 'FileOutputStream'?

➔'FileOutputStream' class is from java.io package and which is instantiated to create a new file with 0-KB and which is linked to write byte – stream – data.

-Syntax:-

FileOutputStream fos = new FileOutputStream("fPath & fName");

36)Define getBlob() methods?

➔**-getBlob()** – method is from ResultSet and which is used to create implementation object for 'Blob' interface.

-This Blob – Object internally connected to the stream – column of ResultSet.

-Syntax:-

```
Blob b = rs.getBlob(3);
```

37)Define getBytes() methods?

➔-getBytes() – method is from 'Blob' interface and which is used to convert stream-data to byte-Array.

-Syntax:-

```
Byte by[] = b.getBytes (1, (int) b.length());
```

38)What is CollableStatement?

➔'CollableStatement' is an interface from java.sql package and which is used to execute Procedure and Functions from Database product.

-We use prepareCall() – method from 'Connecting interface' to create implementation object for CollableStatement interface.

-Syntax:-

```
CollableStatement cs  = con.prepareCall ("{call ProcName/FuncName}");
```

39)What is Function?

➔The part of program which is executed out of main program in C-language is known as Function.

40)What is Member – Function?

➔The Function which can be declared inside and outside the 'class' in c++ is known as Member Function.

41)What is Method?

➔The function which is declared only inside the class in java is known as Method.

42)Define Procedure in SQL?

➔A Procedure in SQL is like a *small program* that you write once and can run (or call) whenever you need it. It is a set of SQL statements (like SELECT, INSERT, UPDATE, DELETE) that are grouped together and given a name.

-Structure in Procedure.

➔create or replace Procedure-name

```
(para_list)
Begin
        query – 1
        query – 2
        ---------
       End;
       /
```

43)Types of Procedure.

➔**a)IN – Parameter Procedure**

The Procedure which take the data from Java program and load on to DB – Product , are Known as IN-Parameter Procedure.

b)OUT – Parameter Procedure

The Procedure which take the data from DB - Product and send to Java Program , are known as OUT – Parameter Procedure.

c)IN – OUT Parameter Procedure

The procedure which perform both operation, which means IN and OUT are known as IN-OUT Parameter Procedure.

44)Define Function in SQL?

➔Function also a set of queries executed at a time on Database Product , and after execution Function will return the value.

-We used 'return' statement to return the value.

45)Structure of Function.

➔-create or replace Function Function-name

```
(para_list) return data-type as var data-type;
begin
        queries
        return var;
        end;
        /
```

46)Define Transaction?

➔Set of statement executed on a single resource or multiple resources using ACID properties is known as Transaction

-A:- Atomicity

The process of executing all statement in transaction at-a-time or not-at-all, is known as Atomicity.

-C:- Consistency

The process in which the selected state – of – resources remain same until the transaction is Completed , is known as Consistency.

-I:- Isolation

The process of executing multiple users independently Is known as Isolation.

-D:- Durability

The process of recording the state of transaction and making it available for user is known as Durability.

47)Define Transaction Management?

➔The process of controlling the transaction from starting to ending, is known as Transaction Management.

-We are used the following method in Transaction Management.

i)getAutoCommit()

ii)setAutoCommit()

iii)setSavepoint()

iv)releaseSavepoint()

v)Commit()

vii)rollback()

48)Define Batch Processing in JDBC?

➔The process of collecting multiple queries into a batch and executed at-a-time on Database Product , is known as Batch Processing in JDBC.

-We use the following methods to perform batch processing.

i)addBatch()

ii)executeBatch()

iii)clearBatch()

49)Define MetaData?

➔The data which is holding information about another data is known as MetaData.

-The following components in JDBC generate MetaData object.

i)Database MetaData

ii)Parameter MataData

iii)ResultSet MataData

50)Which type of Object generated from CoreJava.

➔i)User defined class Object

ii)String – Objects

iii)WrapperClass – Objects

iv)Array – Object

v)Collection<E> Object

vi)Map<K,V> Object

vii)Enum<E> Object

51)Which type of Objects Generated from JDBC?

➔i)Connection Object

ii)Statement Object

iii)PreparedStatement Object

iv)CollableStatement Object

v)ResultSet Object

vi)DatabaseMetaData Object

vii)ParameterMetaData Object

viii)ResultSetMetaData Object

ix)RawSet Object

x)RawsetMetaData Object

52)Define 'Wrapper' in JDBC?

➔'Wrapper' is an interface from java.sql package and which is used to check the components are binded(linked) to Database or not.

-Methods of Wrapper.

i)unwrap():- Give me the real object inside.

ii)inWrapperfor():- Can I get the real object? (Check)

53)Define 'AutoCloseable'?

➔'AutoCloseable' is an interface from java.lang.package and which supports auto-closing Operations in try-with-resource-statement.

-The component which are used in try-with-resource-statement must be implemented or Extended from "java.lang.AutoCloseable" interface.

-JDBC Components like Connection,Statement,PreparedStatement,CollableStatement and ResultSet will supports auto-closing operations, because they are extended from 'AutoCloseable-Interface.

54)Define 'RowSet'?

➔'RowSet' is an interface from javax.sql package and which is also used to hold the result generated from select queries.

-'RowSet' extends from 'ResultSet' , Which means RowSet is a ChildInterface of ResultSet.

-'RowSet' objects are by default Scrollable Objects.

-This 'RowSet' is categorized into two types.

i)JdbcRowSet

ii)CachedRowSet

55)Define JDBC driver?

➔The driver which is used to establish communication between java program and database

Product is known as JDBC driver.

-Types of JDBC driver.

i)JDBC-ODBC bridge driver

ii)Native-API driver

iii)Network protocol driver

iv)Thin driver

56)Java Platforms.

➔-Java Platforms are categorized into the following.

i)Java SE

-Java SE stands for "Java Standard Edition' and, which provide API and environment to develop Non-Server-based-Application(Core Java+JDBC)

ii)Java EE

-Java EE stands for "Java Enterprise Edition" and, which provide API and environment to develop Server-based-Application(Servlet, JSP, JSF, WebServices)

iii)Java ME

-Java ME stands for "Java Micro Edition" and, Which provide API and environmental to develope application related to Machine and Mobile.

iv)JavaFx

-JavaFx will provide API and Environment to develop rich UI pages.

57)Types of Server Based Application.

➔-Server-based-Application are categorized into two types.

i)Web Application

-The Server-based-Application which are constructed using JDBC, Servlet, and JSP are known as Web Application.

ii)Enterprise Application

-The Server-based-Application which are executed in distributed environment and depending on the features like "Security", "Load Balancing" and "Clustering" are known as Enterprise Application.

58)Define Serialization process?

➔The process of Converting object-state into binary-stream is known as Serialization Process.

59)Define DeSerialization process?

➔The process of converting binary stream into object-state is known as DeSerialization Process.

60)Define Serializable Object?

➔The objects which are generated from the classes, which are implemented from 'Serializable' interface are known as Serializable objects(All CoreJava Objects)

61)Define Non-Serializable Object?

➔The objects which are generated from the classes which are not implemented from 'Serializable' Interface are known as Non-Serializable objects(All JDBC Objects)

62)Define Servlet?

➔The Platforms independent Java-Program which is executed in Server environment and interacts with user through Web-Browser, is known as Servlet Program or Server Program.

63)Define Client?

➔Client represents WebBrowser, from where the request is generated for server. The request which is raised from Web Browser is known as HTTP request and which is categorized into two types.

i)POST request:-

-The request which is raised to send the data to server is known as POST request.

ii)GET request:-

-The request which is raised to get the data from the server is known as GET request.

64)Define Server?

➔Server means Service Provider, which means accepting the request and providing the response.

-In realtime Servers are categorized into two types.

i)Web Servers

A Web Server is a software that stores websites and sends web pages to users when they request them through a browser.

ii)Application Servers

An Application Server is a software that runs business logic (like calculations, data processing) and creates dynamic web pages for users.

65)Define Servlet API?

➔ The Servlet API is a set of classes and interfaces in Java that helps you create web applications by handling requests and responses between a web browser and a server.

66)Which are the methods used in 'Servlet'?

➔i)init()

-Init() method is used to perform initialization process which means making the program ready for execution.

ii)service()

-service() method is used to handle the request, which means accepting the request and providing the response.

iii)destroy()

-destroy() method is used to perform closing operation, which means service closing , response closing.

iv)getServletInfo()

-getServletInfo() method is used to write servlet information.

v)getServletConfig():-

-getServletConfig() method is used to write servlet configuration details.

67)Define ServletContext?

➔ ServletContext is an object provided by the web server that gives information about the web application and lets all servlets share data with each other.

68)Define ServletConfig?

➔ ServletConfig is an object that gives information to a specific servlet from the web server.

69)Define ServletRequest?

➔ServletRequest is used to handle the details of the incoming request from the client to the

server in a Java-based web application.

70)Define ServletResponse?

➔ServletResponse is used to build and send a response from the server back to the client in a Java-based web application.

71)Define getParameter() method?

➔ –getParameter() is used to get the data sent from the client (like user input) in an HTTP request.

-Syntax:-

```
String Cid = req.getParameter("cid");
```

72)Define getWriter() method?

➔getWriter() is used to send text data to the client as part of the server's response.

-Syntax:-

```
PrintWriter pw = rs.getWriter();
```

73)Define setContentType() method?

➔setContentType() method is used to specify the type of data that the server is sending back to the client in the response. This helps the browser or the client understand how to process or display the data.

-Syntax:-

```
Res.setContentType("text/html");
```

74)Define 'RequestDispatcher' in Servlet Programming?

➔RequestDispatcher is an interface in Java used in Servlet programming that allows one servlet to forward a request to another resource (like another servlet, JSP, or HTML file) or to include content from another resource in the response.

75)Define JSP Programming?

➔ JSP (JavaServer Pages) is a technology used to create dynamic web pages in Java. It allows you to embed Java code directly into HTML pages, making it easier to build interactive and dynamic websites.

76)Which tag are used to write Java-Code in JSP programs?

➔i)Scripting Tags

ii)Directive Tags

iii)Action Tags

77)Define Scripting Tags?

➔-Scripting Tags are used to write normal basic code in JSP programs.

i)Scriplet Tag:-

-Scriplet tag us used to write direct JavaCode/ServletCode in JSP Programs.

-Syntax

```
<%
    ----------
%>
```

ii)Expression Tag:-

-Expression tag is used to assign the value to variable or send the data to Web Browser directly.

-Syntax:-

```
<%=
    --------Variables,Expression----------
%>
```

iii)Declarative Tag:-

-Declarative tag is used to declare Variable and methods in JSP Programs.

-Syntax:-

```
<%!
    ------------Variables,Methods-----------
%>
```

78)Define Directive Tags?

➔-Directive tag will specify the information about JSP-programs.

i)@page:-

-'@page' tag will provide the information about current running JSP page.

-Syntax:-

<%page language = “java” setContentType = “text/html”.....%>

ii)@include

-‘@include’ tag is used to include the file to the response of Current JSP page.

-Syntax:-

<%@ taglib url = “.....” %>

iii)@taglib:-

-‘@taglib’ tag is used to link external libraries to current JSP page.

-Syntax:-

<%@ taglib url = “.....” %>

79)Define Action Tags?

➔-Action tags will perform some operation while execution process.

-i)<jsp : useBean>

-ii)<jsp : setProperty>

-iii)<jsp : getProperty>

-iv)<jsp : forward>

-v)<jsp : include>

-vi)<jsp : param>

80)Define DAO Layer in MVC?

➔DAO stands for “Data Access Object” and which is separate layer in MVC(Model View Controller) holding database related codes or logics.

81)Define ‘request’?

➔The query which is generated by user through Web Browser is known as ‘request’.

-request is categorized into two types.

i)POST request

ii)GET request

82)Define Session?

➔The time-interval between User-Login to User-Logout is known as ‘Session’.

83)Define Session Management?

➔-The process of recording the state of user and tracking the user in session is known as Session Management.

84)Types of Session Tracking techniques.

➔**i)HttpSession:-**

- HttpSession helps store user-specific data on the server, allowing a web application to "remember" information about the user across different pages and requests.

-Methods of 'HttpSession'

a)public abstract void setAttribute()--The setAttribute() method is used to store data with a specific name, making it easy to access that data later in the same session, request, or application.

b)public abstract java.lang.Object getAttribute()-- In HttpSession, getAttribute() is used to get data (an object) that was saved in the user's session earlier.

c)public abstract void removeAttribute()-- In HttpSession, removeAttribute() is used to delete something that was saved in the session. It removes an attribute (data) by its name (key).

d)public abstract java.util.Enumeration<java.lang.String> getAttributeNames()-- In HttpSession, Enumeration is used to list or go through all the attribute names that are stored in the session.

e)public abstract void invalidate()-- In HttpSession, invalidate() is used to destroy or end the user's session completely.

85)What is URL re-Write?

➔The process of adding parameter value to servlet-url-pattern is known as 'URL-re-Write' process.

-The process is used to send some data from one servlet-program to another servlet-program in Session – Tracking process.

-Syntax:-

Servlet-url-pattern?para1 = value&para2 = value&para3 = value&……..

86)What is Hidden Form Fields?

➔The process of declaring <input type = "hidden"---> in <form> tag is known as Hidden form fields.

-Syntax:-

```
<form action = “url” method = “POST”>
<input type = “hidden” name = “nm” value = “v”>
----------
</form>
```

86)What is Cookie?

➔ A cookie is a small piece of data that a website stores on your browser (your computer or phone). It helps the website remember you when you visit again!

87)Types of Cookies.

➔**i)Persistent Cookie:-**

The cookies which are stored and available in Web-Browser until user logout , are known as Persistent Cookies.

ii)Non-Persistent Cookie:-

The cookies which are destroyed automatically when the Web-browser is closed , are known as Non-Persistent Cookies.

88)Which are the methods used in Cookies?

➔**i)setMaxAge(int):-** setMaxAge(int) is used to set how long the cookie should stay saved in the browser.

ii)getMaxAge():- getMaxAge() is used to find out how long (in seconds) the cookie is supposed to live.

iii)setValue():- setValue() is used to set or change the value of a cookie.

iv)getValue():- getValue() is used to retrieve the value stored in a cookie.

89)Execution behaviour of addCookie() method.

➔addCookie() method will add cookie to response-object, In this process it will perform Serialization process.

-Syntax:-

Res.addCookie(ck);

90)Execution behaviour of getCookies() method.

➔getCookies() method will count the cookies from the request – object and generate array of size

equal to the number of cookies , in this process it also reconstruct. (deserialization) the cookie – object and add to the array.

-Syntax:-

Cookie c[] = req.getCookie();

91)What is the difference between HttpSession and Cookie?

➔i)'HttpSession' is an interface , and 'Cookie' is class.

ii)HttpSession is server dependent , and Cookie is Client(WebBrowser) dependent.

iii)HttpSession is more secure than cookie.

92)What is the difference between getSession() , getSession(false) , and getSession(true)?

➔**i)getSession():-** getSession() method will look for existing session , if available access the existing Session , if not available create new session.

ii)getSession(false):- getSession(false) – method will look for existing session , if available access the existing session , if not available new session is not created.

iii)getSession(false):- getSession() method will look for existing session , if available access the existing session , if not available create new session.

93)What is the difference between JAR , WAR , EAR?

➔**i)JAR:-** JAR stands for 'Java Archieve' and which is Compressed format of more number of class files.

ii)WAR:- WAR stands for 'Web Archieve' and Which is Compressed format of Html files , Xml files , Servlet , JSP , JAR's and other UI files.

iii)EAR:- EAR stands for 'Enterprise Archieve' and which is Compressed format for JARs , WARs and other external services.

94)What is Web Application Architecture Models?

➔The design models which are used to construct Web Application are known as 'Web Application Architecture Models'.

-These Web Application Architecture Models are categorized into two types.

i)Model – 1 Architecture:- Model-1 is the first architecture to develop Web Application , and Which is JSP is the controller of total application.

ii)Model – 2 Architecture:- Model – 2 Architecture depends of MVC Architecture.

95)What is MVC?

➔**-M:-**Model Layer represents Bean – objects , where the data is available , Model layer also Included JCF , DAO and other Services.

-V:-View Layer represents Presentation and in this layer we use JSP.

-C:-Controller represents accepting the request from user and controlling all layers in the application.